

# Yucheng (Cecilia) Zhang

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## EDUCATION

### Northwestern University

*M.S. in Computer Science*

Aug. 2026 – May 2028

Evanston, IL

### The University of Texas at Austin

*B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00*

Aug. 2023 – May 2026

Austin, TX

**Relevant Coursework:** Stochastic Processes, Probability and Statistics, Real Analysis, Data Analytics, Database Systems, Artificial Intelligence

## TECHNICAL SKILLS

**Analytics/Modeling:** Credit risk, predictive modeling, statistical analysis, feature engineering, model validation, backtesting, ROC/AUC

**Languages/Tools:** Python, SQL, R, MATLAB, SAS, Power BI, Git, Excel (reconciliation, validation, reporting)

**Data Platforms:** SQL Server, Oracle, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker

**Financial Applications:** Credit risk modeling, fraud analytics, transaction monitoring, SEC filing analysis, customer risk scoring

## RISK ANALYTICS AND FINANCIAL DATA EXPERIENCE

### Enterprise Risk Analyst Intern

May 2025 – Dec. 2025

*Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI*

Dallas, TX

- Built an end-to-end credit risk modeling pipeline from raw SQL Server data, spanning ETL, credit-feature engineering, Logistic Regression, Random Forest training, and validation.
- Engineered credit risk features including PayNet inquiries, loan-to-value ratios, account tenure, asset segments, and loan servicing attributes for trucks, trailers, and used-vehicle portfolios.
- Validated model performance with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by approximately **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard segmentation updates.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by 10% and supporting recurring portfolio risk monitoring.

### Financial Data Development Engineer – Fraud Detection

Jul. 2024 – Aug. 2024

*Caitong Securities Ltd | SQL, Oracle, Stored Procedures*

Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% in transaction-monitoring workflows.
- Developed a customer risk scoring engine by integrating transactional and behavioral data streams for fraud analytics, reviewer triage, and AML-related review workflows.

### FinChat – AI-Powered Financial Analysis System

Jan. 2023 – May 2024

*Southern Methodist University | Python, FinGPT, LangChain, CrewAI*

Remote

- Built ETL pipelines to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial data into a scalable data lake for metric extraction and trend detection.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving data processing accuracy by 3% and reducing manual review time for filing analysis.
- Collaborated with a cross-university team to connect FinGPT-based retrieval, filing preprocessing, and financial-question workflows into a reusable analysis prototype.

### Software Engineering Intern – Customer Analytics

Nov. 2023 – Jun. 2024

*Joblogic-X Corporation | Python, APIs, Git*

Dallas, TX

- Integrated social-media and transactional order data into a customer analytics pipeline for ranking customers, segmenting engagement patterns, and supporting marketing strategy.
- Implemented Python ranking and clustering algorithms, producing prioritized customer lists and reproducible analytics outputs for downstream chatbot workflows.

## ADDITIONAL QUANTITATIVE RESEARCH

### Researcher & TA – Biological Data Analytics

Jan. 2024 – Dec. 2024

*University of Texas at Austin | R, Linux, AWS, Docker*

Austin, TX

- Processed **250GB** of RNA-seq data with reproducible R/Linux workflows, performing Voom/Limma differential-expression analysis and identifying 48 key genes.
- Automated normalization, PCA, and volcano-plot reporting workflows, improving processing speed by 6% and interpretation efficiency by 15%.
- Applied transferable statistical methods including feature extraction, time-series comparison, permutation-based FDR, and data-quality validation across large datasets.

## SELECTED RESEARCH AND PUBLICATIONS

- Coauthored **5 peer-reviewed publications** applying machine learning and statistical analysis to computer vision, neuroengineering, and social-media data.
- Selected venues include *Expert Systems with Applications*, *Frontiers in Neurology*, *Frontiers in Neuroscience*, and IEEE ICIPCA, with first-author work in neuroengineering and EEG-based emotion recognition.